Open-Source Development

ITIC2262

Presented by:
Lieutenant Commander (IT) RMJP Nayanajith

Open OS Configurations

1. Simple Linux Commands

- Date - Display Current date
- Cal - Display Calendar
- whoami - Display Current account
- who - Command in Linux is a command-line utility used to display information about users currently logged in to the system. It provides details about each active session, including:
  - **Login Name:** The username of the logged-in user.
  - **Terminal (TTY):** The terminal or pseudo-terminal (PTS) from which the user is logged in.
  - **Login Time:** The date and time when the user logged in.
  - **Remote Hostname (if applicable):** If the user is logged in from a remote machine, the hostname or IP address of that machine is displayed.

- More -

  The more command in Linux is a utility used to view the contents of text files one screenful at a time.

  - **more anaconda-ks.cfg**

- Less -

  The less command in Linux is a powerful utility for viewing the contents of text files or the output of commands, one page (or screen) at a time

  - **less anaconda-ks.cfg**

- Tail - To view the last 10 lines of a file

  - **tail anaconda-ks.cfg**

Head

- The head command in Linux is a utility used to display the beginning of a file or the output of a command.

- By default, it displays the first 10 lines of the input.
  - **head anaconda-ks.cfg**

Last -

To view a chronological list of recent user logins and logouts, simply execute the command without any arguments

- **last**

History –

The history command in Linux displays a list of previously executed commands in the terminal.

- **History**

2. Help Commands

- Man : To view the manual page for a specific command, use the following syntax.
  - **man ls**

- Whatis : Displays a one-line manual page description for a specified keyword or command
  - **Whatis mkdir**

- Info : a utility used to read multi-page, hypertext documentation
  - **info date**

- -help
  - **fdisk -help**

3. File and Directory Management

- **ls - Lists the contents of a directory**

  - `ls -l` : long format with details
  - `ls -a` : show hidden files
  - `ls -la` : Combination of above Commands
  - `ls -lh` : (human-readable) option to display file sizes in a more readable format
  - `ls -lr` : (revers) option to list/display file in reverse order,

- **cd** - Changes the current working directory.
  - `cd var/www/42/ict : go to a specific directory`
  - `cd .. : go up one directory`
  - `cd ~ : go to home directory`
- **pwd** - Changes the current working directory.
- **mkdir** - Creates a new directory.
  - `mkdir : mydirectory`
  - `mkdir -p /path/to/new/directory : create parent directories if they don't exist`

- **rmdir**: Removes an empty directory.

- **rm** : Removes files or directories.
  - `rm myfile.txt`
  - `rm -rf mydirectory`
    : recursively remove a directory and its contents

- **cp** : **Copies files or directories.**
  - `cp source.txt destination.txt`
  - `cp -r sourcedir/ destinationdir/`
    : recursively copy a directory

- **mv** : Moves or renames files or directories.
  - `mv oldname.txt newname.txt`
  - `mv file.txt /path/to/new/location/`

- **Touch:** Creates an empty file or updates the timestamp of an existing file.
  - `touch newfile.txt`
- **Cat : Displays the content of a file.**
  - `cat newfile.txt`

3. System Information and Monitoring

- **Uname** : Displays system information
  - `uname -a : all system information`
- **Hostname** : Shows or sets the system's hostname.
- **Top** : Displays real-time information about running processes and system resources.
- **Df** : Shows disk space usage.
  - `df -h : human-readable format`
- **Du** :Displays the disk usage for the specified file or directory.
  - `du <file_or_directory>`
- **Runlevel** **:** Display runlevels 0 – 6 levels
- **Free** **:** Shows memory usage.
  - `free -h : human-readable format`

- **ps** : Displays information about running processes.
- **uptime**: Shows how long the system has been running.

4. Hardware Information Commands

➢**lscpu**	See **CPU** information.
➢**lsblk**	See information about **block devices**.
➢**lspci -tv**	**Show PCI devices** (graphics card, network card, etc.) in a tree-like diagram.
➢**lsusb -tv**	Display USB devices in a tree-like diagram.
➢**cat /proc/cpuinfo**	Show detailed CPU information.
➢**cat /proc/meminfo**	View detailed system memory information.
➢**cat /proc/mounts**	See mounted **file systems**.
➢**free -h**	Display **free and used memory**.
➢**sudo dmidecode**	Show hardware information from the **BIOS**.

5. Opening and Creating Files

- Vi/Vim

  - To open an existing file

    vim filename.txt

  - To create a new file.

    vim kdu.txt

- Insert Mode:

  - Used for entering and editing text.

    - Press **i** to insert text **before the cursor**.
    - Press **a** to append **text after the cursor**.
    - Press **o** to open a new **line below** the current line and enter Insert mode.
    - Press **O** to open a new **line above** the current line and enter Insert mode.

- Saving and Quitting (in Command-Line Mode): Press Esc
  - :w (save)
  - :q (quit)
  - :wq or :x (save and quit)
  - :q! (quit without saving, discarding changes)

6. Linux Based Net Configurations

1. **Identify the Network Interface –**
   Linux command to identify the available network interface(s).

   Command: ip link

2. **Configure an IP Address –**
   Configure the network interface with:
   IP Address: 192.168.10.10
   Subnet Mask: 255.255.255.0 (/24)

   Command: sudo ip addr add 192.168.10.10/24 dev eth0

3. **Verify the Configuration –**
   Display and verify the assigned IP address.

   Command: ip addr show

4. **Test Network Connectivity**
   Test connectivity using the ping command.

   Command: ping -c 4 192.168.10.1

Logging

* User Levels
  * Super user (#)
  * Limited Users ($)
* Local Logging
  * Su
  * Su –
* Remote Logging
  * Ssh
  * root@

Thank You